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Illumi's world: A mini-game development with parametric BIM-based simulations

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Abstract

The research investigates the design and development of a serious game to teach green building design and energy literacy in rural middle schools in the United States. The paper presents a pilot study, education mini-game development integrated with parametric BIM and energy simulations. The game scenario was built on the developed science curriculum modules in our funded research, teaching building energy technologies such as daylighting, artificial lighting, window configurations, building materials, solar panels, etc. The mini-game, *Illumi's World*, presents a baseline science lab and a media library of typical public schools in the United States. The players have the opportunity to improve energy literacy in several ways: manipulating the building configurations and the energy options, reviewing energy costs and emission level changes, and monitoring the performance from the game dashboards. This paper presents background theory, curriculum design, the mini-game development framework, methods and tools for energy simulation and BIM visualization, and the findings and challenges.

Keywords

Game-based learning, green building literacy, systems thinking, model-based reasoning, building information modeling, energy simulation

Context of game-based learning for green building education

The contemporary green building movement seeks to promote buildings that lessen environmental impacts through better building construction, building operation and maintenance, and lifecycle cost.[1,2] A primary focus of green building design is energy performance, which includes energy sourcing and the reduction of greenhouse gas emissions in the built environment. While several organizations have crafted definitions for a

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green building (e.g.,[3–5] all approaches prioritize high-performance building design that approaches net zero (US EPA, 2020). "Net zero" refers to high-performing buildings that generate as much energy as they consume.

Much science content underlies the how and why of net-zero building design; however, net-zero green building ideas have rarely been integrated into formal science classrooms.[6,7] There are broader efforts to enhance youth energy literacy, but the teaching apparatus and curriculums of formal education in the United States still do not reflect these global demands.[8]

The research is built on the theoretical foundation of systems thinking and model-based reasoning (MBR). To understand energy sourcing, flow, and use within the built environment, students need systems thinking abilities.[9] Systems thinkers can look at individual parts and processes of a system and understand how these individual elements both fit within and affect the system as a whole.[10] Within systems thinking, students can distinguish and identify system organization and analyze system behavior.[11] One of the primary methods of teaching systems thinking is MBR, where large complex systems are split into small manageable components.[12] The result of MBR to teach systems thinking, however, depends on how instructional scaffolds support learners to understand how the smaller manageable components fit back into the larger system. It presents a unique challenge for instructors, especially when learners present difficulties in making sense of the material.

Game-based learning and energy modeling/simulation situated within MBR are promising vehicles to support students in systems thinking.[13] Digital tools in education engage learners with complex systems in more meaningful ways, as students can manipulate causal factors to figure out what happens at different time scales and systems levels.[14] Using these tools improves students' knowledge gains[15] and can enhance learning effectiveness.

Game-based learning, also known as serious games, refers to games designed to enhance knowledge or for training to improve skills rather than simply for entertainment.[16] Serious games take the concept of video game technologies and repurpose the game for training, education, advertising, national defense, and more.[15] In addition, serious games show promise for learning through informal means of creativity,[17] play,[18] collaboration,[19] and role-playing.[20] Studies describe the benefit of serious games in comparison with the conventional education media, including intrinsic motivations, greater attention, enhancing the sense of engagement, lowering the threat of failure and mistakes, application of learned skills, greater intellectual intensity, and more.[21–23]

During the past decades, various serious games were developed as a medium to improve energy literacy in the built environment, such as Efficient City (2001), ElectroCity (2007), Clim'way (2010), CityOne (2011), EnerCities (2011), EnergyVille (2011), and Dropoly (2010). ElectroCity challenged gamers to balance energy consumption, development revenue, and renewable energy generation in town planning and design. Clim'way, developed by UC Santa Barbara, asked users to monitor and reduce the energy consumption of the community plan during a 50-year life cycle. The game challenges users to reduce greenhouse gas emissions by 75 percent and increase renewable energy production by 60 percent throughout the building life cycle. Energy Efficiency Game by National Energy Foundation is a web-based 2D game asking gamers to reduce their energy use at home. More recently, serious game research addressed behavior change of the specific end-user, widely accepted as the most impactful factor for energy savings. For example, Serena Supergreen (2018) was designed to motivate and educate female students between 12 and 16. Powersaver Game (2019) challenged virtual family members to save 15% on energy, electricity, and gas consumption in 3 weeks.[24] claimed that these serious game developments show the advantages, including better visualization of energy consumption, increased end-user engagement, their energy behavior change, and collectible end-user feedback.

Serious games provide an innovative approach to computer-based modeling to support the development of systems thinking.[25] However, many unknowns remain as to how games can help reduce the cognitive burden

of novices through systems thinking.[13] Despite rapid research and development of serious game technologies, these educational technologies are still poorly interwoven with science curriculum units. Energy data in the game does not interact with various energy options and game scenarios. Since[26] claimed the potential of CAAD and serious game integration in architectural design education, a growing number of studies have explored BIM and game integration for visualization, design decision-making, design education, professional training, etc. For example, BIM Game[27] explored the potential benefits of BIM-based serious game development to educate non-design experts, focusing on energy efficiency and sustainability.[28] claimed that integration of BIM, IoT, and serious game would improve the learning environment in the construction industry. There have been several attempts to improve the education and training of professionals in the Architecture, Engineering, and Construction (AEC) industry, but it is still in its infancy. Our multidisciplinary research team from education, education technology, and building science envisioned that the advancement of BIM, increasing visualization capabilities, accessible game developments, and multicriteria energy simulations would boost the efforts in this field.

Mini-game development framework

The research carried out a pilot study, education mini-game development integrated with parametric BIM and energy simulations. The goal is to create a theoretical and empirical framework towards NetZero Game, a large-scale serious game that teaches green building design and energy literacy in middle school classes. This section describes background theory, the developed curriculum, the mini-game development framework, and methods and tools for energy simulation and visualization. The target application is the rural school districts in Missouri, underserved populations for science, technology, engineering, mathematics (STEM) interventions with limited access to STEM learning opportunities.

Model-based reasoning and EYE curriculum

Prior work has focused on the numerous energy misunderstandings students have about energy systems.[29] Research outcomes suggest that teaching energy systems should make energy transfer and transformation visible, observable, and manipulable.[30] We contend this occurs through MBR.[31,32] We define MBR as students' development, use, evaluation, and revision of their own models as thinking tools for causal interactions within systems.[33] Students develop a model to answer a question or consider a problem about elements and their relationships within a system. Models can be both drawings, writings, or other products that exist in the digital realm, analog, or both.[34] The initial model is built on students' prior knowledge in which students consider the components of the system (elements, objects, and entities) and how these components causally interact to form system behavior.[35] As student knowledge about the system grows, they evaluate this model for how well it supports their system reasoning and revise their model based on their new understanding. As students revise their model, they can choose the cause-and-effect relationships within the energy system to amplify while reducing system complexity through backgrounding other system elements.[31]

MBR also serves as a learning scaffold for students' energy learning. Scaffolds are learning environment structures that support students in learning about complex systems in ways they would be unable to do in the absence of such a support.[36,37] The model serves as a scaffold as it is a concrete representation of an abstract idea. Students are able to make system elements (components, relationships, causal mechanisms) visible to advance their reasoning about system behavior. The model also makes visible student knowledge gaps about system components and relationships which prompts them to ask questions, seek out new ideas, and supports creating cognitive links about new ideas about energy systems within their conceptual understanding.

To support students in conceptualizing an energy system, we embedded MBR within a six-lesson curriculum unit called Energy and Your Environment (EYE), funded by the NSF (See Table 1). EYE fosters

Table 1. EYE Unit lessons.

Lesson	Description
Lesson 1: What is energy?	Introduce energy and energy flow within the school building
Lesson 2: How do humans harness energy?	Introduce energy resources and how these energy resources become different forms of energy (potential and kinetic energy) used by the school building
Lesson 3: What are energy harnessing impacts on Earth systems?	Activities on building-related energy technologies following energy from resource to energy use focusing on the transformation and transfer of energy within the school building
Lesson 4: Where is light energy within the school building?	Examine energy flow within natural and artificial lighting systems in the school building
Lesson 5: Where is thermal energy within the school building?	Examine energy flow within HVAC systems in the school building
Lesson 6: How can you design and build an energy-efficient school building?	Design and build a one-room schoolhouse using energy-efficient ideas

place-based education by using local school buildings to enhance systems thinking about energy consumption and flow between buildings and the natural environment. The EYE unit represents an interdisciplinary approach across science, architectural design, and a technology-supported engineering design process motivating students' energy literacy. The unit was designed to align with the Next Generation Science Standards,[38] one of the national science education standards in the United States.

The mini-game was embedded within two selected EYE lessons: lessons four and five. Lesson four focused on light energy, aligned with the middle school NGSS standard for light waves. Students should show proficiency in understanding how light is reflected, absorbed, and transmitted through different materials depending on the type of lights and materials. Lesson five focused on thermal energy in which the NGSS standard for thermal energy. Students need to show proficiency in light energy transformation and energy transfer by convection, conduction, and radiation. The game uses MBR to scaffold students' knowledge building about energy use and flow within the built environment.

Process

Before creating our intended full-scale 3D serious Net-Zero game, the research explored and examined a 2.5D mini-game, *Illumi's World*. The mini-game enabled us to understand and facilitate how we communicate between the curriculum, the simulation, and the game elements. A 2.5D game uses fixed isometric views to enable 2D gameplay in a 3D environment. Players can move throughout the game environment, but the camera view does not change until a new game scene is accessed. The game scenario was built on lessons four and five in the EYE curriculum, teaching energy technologies such as daylighting, artificial lighting, window configurations, building materials, solar panels, etc. *Illumi's World* presents a baseline science lab and a media library of the typical public schools in the United States. The players can have the opportunity to improve energy literacy in several ways: manipulating the building configurations and the energy options, reviewing energy cost and the emission level changes, receiving interactive feedback from a game dashboard and virtual consultants, and iterating the changes to achieve energy goals.

The game prototype was developed in three phases, including (i) EYE curriculum development (C1, C2), (ii) Game design and development (G1, G2, G3), and (iii) Energy modeling and simulation (S1, S2, S3) shown in Figure 1.

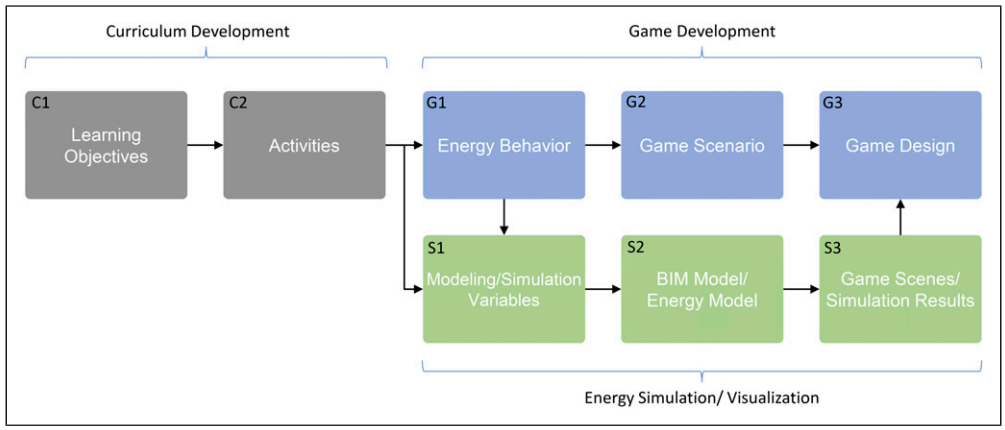


Figure 1. Research phases of mini-game development.

First, parametric BIM was used to create manipulatable virtual models, such as changing ceiling heights, exterior wall configurations, window-to-wall ratio, shading devices, and artificial lighting configurations. Then, the real-time visualization platform converted the parametric BIM into a series of game scenes. The research explored Unreal Engine's Twinmotion and Enscape 3D. These visualization platforms eliminated a time and effort-intensive conversion process between BIM and game visualization. For energy simulations, the research used BIM-based cloud simulation that runs EnergyPlus engine. The cloud simulation generates the energy model, provides benchmark data of energy savings based on the building configuration changes, and visualizes the simulation results for straightforward understanding for the broad stakeholders. After the simulation runs, the simulation results and the benchmark data were aligned with the energy options in the game. The simulation results include Energy Use Intensity (EUI), energy cost, and CO2 emission levels. For the game development, we used Unity3D and several game assets. In addition, the research created custom creature characters for better student engagement in a fictitious science lab. The creature enabled learners to make decisions throughout the game, controlling the lab environment to improve the creature's health conditions. In the game scenario, the creature's health is sensitive to the building energy performance, indoor daylighting quality, and environmental footprints.

Although the figure visualizes a linear and top-down process sequentially, the research was built on bidirectional data exchange, frequent feedback loops, and iterative synchronizations across three domains of curriculum design, game design, and BIM-based modeling/simulations.

Building modeling and visualization methods

The mini-game development attempted to take advantage of parametric BIM for rapid model creation, iterative manipulation of the model configuration, and reliable synchronization between the building model, energy simulation, and game visualization. In the scope of the full serious game development, the number of building/energy models could incrementally increase when the Game allows players to manipulate building configurations and energy choices. Such an extensive number of models makes the manual creation of energy models and game scenes a less viable approach.

Architecture design and construction domains rigorously investigate BIM and visualization integration. The relevant immerging techniques to this research include real-time AR/VR creation from BIM, pushing BIM data to visualization interfaces, and bidirectional model synchronization between BIM and

visualizations. Ideally, these features would give students various learning experiences: students can variate a wide range of energy options in BIM; they can monitor how their choices change the indoor environment; and students can access building information such as materials, dimensions, and other energy-related building parameters within BIM. In that regard, the research has investigated and benchmarked the feasibility and applicability of BIM-based real-time Visualization and BIM-game integration.

Energy simulation methods

Towards serious game development, the research investigated available integration methods of game and energy simulation. The research compared three integration methods of (i) pre-simulation, (ii) iterative parametric simulation, (iii) and algorithm-based prediction in Figure 2. Three approaches vary in view of process time, feedback latency, maintainability, and dissemination reliability.

Three integration methods take different approaches in the energy model creation from building design, resulting in different accuracy, energy modeling process, and accuracies. Our observations are as follows:

Pre-simulation. Complete simulations first and pass the simulation results to the game development phase. It is possible to use legacy-simulation engines with rich validity and accuracy in results, but time and effort-intensive input processing is required. In addition, the complete list of energy simulation variables and model configurations should be determined before the game development. The research explored BIM-based cloud simulations to eliminate redundancies and errors in energy modeling in legacy simulation engines. Benchmark data from Autodesk Insight 360 enabled the prediction of various energy scenarios. Each simulation run needs to update the energy simulation configuration from BIM models and the simulation interfaces.

Iterative parametric simulation. Automate the energy model update and simulation execution using parameter iteration. Software prototypes could automate the conversion from building models to energy models, parameter updates, simulation runs, and result aggregation. Energy simulation variables and model parameters need to be fully identified before the simulation runs like Pre-simulation. Our prior works examined iterative parametric simulation using BIM, multi-criteria simulation interfaces, and legacy simulation engines.[39] The research used *Autodesk Revit* for building and energy modeling, *Rhino/Grasshopper* simulation plus-ins such as *Ladybug* and *Honeybee* tools, *TT Toolbox* for the iterator, and *Rhino.Inside* for the crossover data communication.

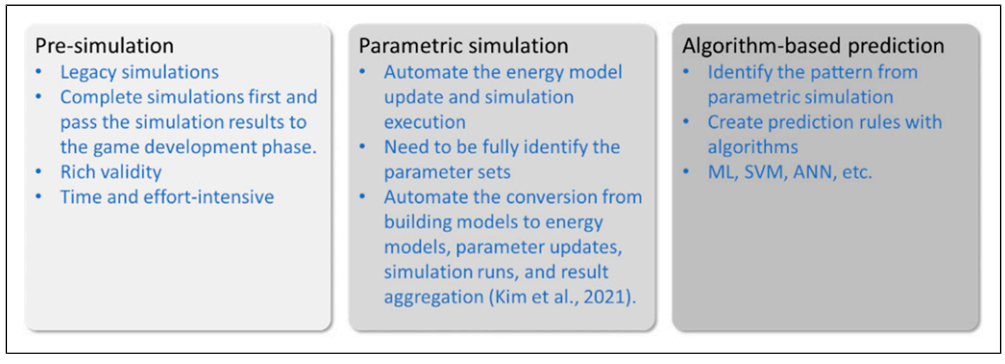


Figure 2. Integration methods of curriculum development, energy simulation, and game development.

Algorithm-based prediction. Identify the pattern between energy simulation variables and the energy performances using AI algorithms. In this approach, prediction models are created using the large number of simulation results obtained from the iterative parametric simulation and Machine Learning (ML) algorithms. After the prediction model creation, it can provide the energy performance projections without using the pre-simulation and the iterative parametric simulation. Our ongoing research[40] investigates cross-platform integrations using BIM for energy modeling, iterative parametric simulation for training and testing data for algorithms, and ML algorithms for energy performance predictions.

The mini-game in this paper presents our first implementation of pre-simulation using selected curriculum modules, BIM-based cloud simulations, and BIM-based visualizations.

Mini-game development

To enable us to better establish a workflow between the three domains, we investigated a smaller section of the Net-Zero Game, a mini-game called *Illumi's World*. Focusing on the smaller-game allowed us to explore sub-lecture module developments from the EYE curriculum. Our major intention in *Illumi's World* was to understand how to best challenge students through specific game activities that improve energy knowledge gain. The following sections explain the research phases illustrated in Figure 1.

EYE learning objectives and activities

The first phase was critical to align the curriculum contents, game scenarios, and building/energy modeling. Table 2 shows examples of essential activities of EYE that provided a foundational framework for game scenarios and energy modeling/simulation (C1 and C2 in Figure 1). The curriculum was initially designed to use analog teaching tools for rural schools where education technology is inaccessible. The mini-game is an intervention to substitute these analog tools with interactive digital learning platforms. The research identified a series of energy behavior from these curriculum contents and relevant activities such as building material change, window configuration change, and lighting fixture change.

Energy behavior and simulation variables

Next, we formulated energy behavior and simulation variables from the learning objectives and activities in the EYE unit. This process is crucial to identify the constraints and conditions for the game scenario development (G2 in Figure 1) and the energy modeling and simulations (S1 in Figure 1). Table 3 shows the matrix between EYE curriculum modules, energy behavior in the mini-game, and energy simulation variables.

The energy behavior in the game is associated with the variables of energy models and simulation in complex ways. For instance, repositioning and resizing rooms in the game changes the dimensions of building surfaces, the orientation of each surface, the level of solar and daylighting gains, the volume of heating and cooling, and many other simulation variables. Changing window sizes impacts the opaque wall size, the window-to-wall ratio, the solar radiation gains through windows, the indoor illuminance level, air infiltration through windows, etc. In most cases, it was observed that there are one-to-many associations between energy behavior and simulation variables, making this mapping process complex. In the classroom setting, such complexities between energy behavior in the game and the simulation variables would be the critical content to understand the impact of their energy choices in building on the energy performances.

Table 2. Main module's activities and learning objectives.

[Module] "Activity"	Learning objectives
[Sun and building] "Light Where You Live" Use climate simulations to examine how sunlight interacts with your school building in your unique location	Apply knowledge of the sun path and solar accessibility, considering building siting and façade configurations
[Building operation] "How you use your building" Change building operation of HVAC, lighting, and other electric equipment. Then observe changes in overall energy use level	Recognize the impact of building occupancy and operation on energy use
[Building materials] "Light & Surfaces" Predict and test how light will interact with various building materials. Describe the ways in which the waves are reflected, absorbed, or transmitted through various materials	Apply knowledge of lighting travel based on building materials of walls, ceilings, floors, and windows
[Daylighting] "Light & Your School Building" Apply what you have learned about the characteristics of light energy and light energy transfer to your school building. As you tour the building, make observations about how light travels inside	Apply understanding of climate data to analyze how light interacts with the school building
[Artificial lighting] "Measuring Light" Use light meter to observe changes in the amount of light in the classroom throughout 2 days, examining the need for artificial light to supplement the natural light that comes into the classroom	Recognize the ability of daylight to reduce energy demand from artificial lighting
[Artificial lighting] "Which Lightbulb?" Students measure the heat output and illumination of different lightbulbs, making and testing their hypotheses about lighting, heat, energy use, and cost over time	Develop an artificial lighting plan that balances the lighting quality and energy cost

Table 3. Energy behavior and simulation variables.

Curriculum modules	Energy behavior in game	Simulation variables
[Sun and building]	Reposition the buildings	Orientation
	Resize the rooms	Energy model dimension
[Building materials]	Select roof materials	R-value, U-value
	Select wall materials	
[Building operation]	Select occupancy schedule	Occupant heat gain
	Select equipment efficiencies	Equipment heat gain
[Daylighting]	Select window size	Window-to-wall ratio
	Select window numbers	R-value, U-value
	Select glass materials	Window transmission
	Select shading devices	Infiltration rate
	Select window opening frequency	
[Artificial lighting]	Select lighting types	Lighting efficiency
	Select lighting efficiencies	

Parametric modeling and energy simulation

The energy behavior and simulation variables provided criteria of parametric modeling in BIM for game visualization, energy modeling variables, and energy simulation scenarios. In *Illumi's World*, the research coincided with game design and game scenario development and had frequent feedback loops to align the game, energy simulation, and game visualization. The research tested the following simulation variables in Table 4.

The research performed 243 pre-simulations, calculated EUI, energy cost, and CO2 emission levels, and fed the simulation results into the game interface. To match the simulation set and the game scenario, we assigned simulation id numbers to the simulation results. The simulations used Autodesk Revit and Insight 360. The visualization was based on BIM-based real-time visualization plug-ins, including Enscape and Unreal engine's Twin motion.

Game design and scenario

The game design was informed by entertainment games popular with middle school-aged students learned through a survey of middle schoolers. Games, such as Pokémon, which involve the process of evolving a creature through various actions, played a role in the larger story development. The social mechanic used for decision-making to design the lab space was influenced by popular gaming spaces such as Roblox. Building on insights from middle school students and their preferences, the mini-game scenario and interface design aimed to enrich engagement and motivation in learning through various game elements.

The core dynamic of the game is creating and building, where players' actions include design decisions that further develop a fictitious lab building.[41] Specifically, the goal of Illumi's World is to raise a creature known as an Illumi through design decisions for the indoor habitat where the creature will either thrive or perish based on the player's energy choices. *Illumi's World* takes place in a research facility on an isolated island in the Galapagos, where the player can interact with other characters. Players assume the role of a scientist tasked with caring for evolving this new species of creatures that are sensitive to climate change (Figure 3).

Other characters include a director of the research station, responsible for the overall energy facility and cost. A lab assistant assists the player with making decisions and finalizing results at the end of each game day. A librarian who provides additional information about related energy subjects. And an engineer who helps the player make critical decisions about energy technology and options. The game environment consists of a multi-building research facility that generates its own power through various means that the player can

Table 4. Simulation category and variables.

Category	Variable types		
Wall/Roof materials	High performance (R38)	Moderate performance (R20 to 25)	Low performance (Uninsulated)
Window-to-wall ratio	Large window (80%)	Medium window (40%)	No window (0%)
Window shades	Large shading (1/2 window height)	Medium shading (1/4 window height)	No shading devices
Glass materials	High performance (Triple Low-E glass)	Moderate performance (Double clear glass)	Low performance (Single clear glass)
Lighting efficiency	High performance (LED)	Moderate performance (Fluorescent bulb)	Low performance (Incandescent bulb)

manipulate. The facility consists of five different spaces: a nursery lab, library, staff lounge, research center, and an observation deck.

The pilot game was developed using Unity3d with several purchased assets and tools to enable us to focus on how to connect the simulation data to the game. The game development started using the 2.5 d Toolkit to create fixed isometric views and facilitate player movements around the isometric environments. It eliminated the extra coding process of camera position and angle adjustment. The 2.5 d Toolkit environments was combined with another tool called Dialogue System for Unity which would enable players to interact with the other characters and make decisions through dialogue options (Figure 4). For character modeling, we used our creature design and some purchased 3D character models for scientists, engineers, and librarians. With our own isometric view images created from Revit models, we were able to assemble our prototype. For feedback in game-based learning, we implemented visual feedback and an in-game tablet, aligned with previous research precedents on feedback methods in the game and simulation-based learning environments.[42] To provide visual feedback, the game interface swapped a series of isometric views created from our Revit

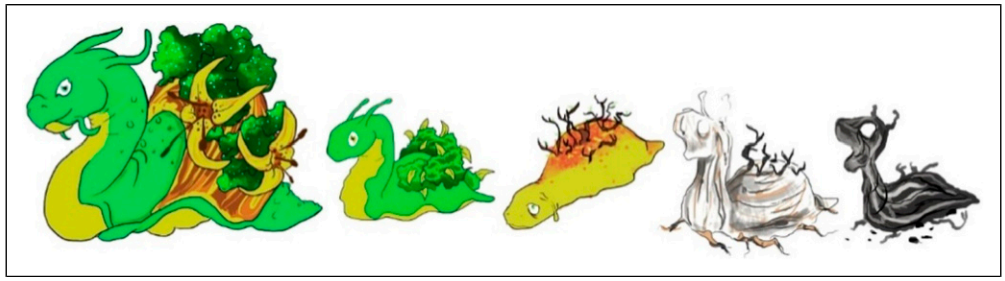


Figure 3. Creature design with five health levels.

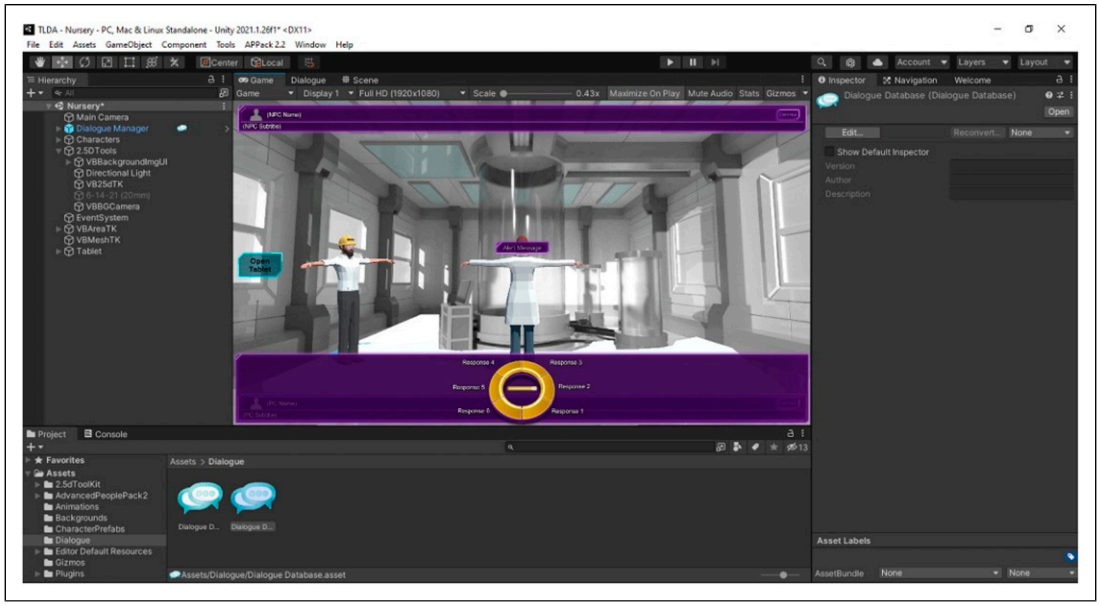


Figure 4. The lab scene development in Unity 3D.

models and visualization platforms according to the selection of energy options. We additionally created a formative and summative feedback performance system through the in-game tablet where players could review their choices and the resulting outcome on the creature's health, EUI, and greenhouse gas emission levels.

Results and discussion

The research investigated mini-game development for rural middle schools based on our energy curriculum, including game scenarios, characters, dialogues, and game interfaces. The research also explored BIM-based parametric modeling and interactive visualizations to expedite game scene creation. BIM-based cloud simulation was carried out for energy modeling and simulation result creation. Figure 5 shows the resulting game prototype of *Illumi's World*. The game contained several elements that framed the game scenario and how players would interact with the data and the game's interface. The top and bottom of the screen display dialogue boxes, providing players with game introductions and guidance based on the game dialogues in Table 5. The players can move the characters in isometric views, read their dialogues, make changes in the energy options, and monitor the game's progress from the in-game tablet.

The game begins in the main lab where the player is introduced to the creature, the lab assistant, and the tablet for receiving feedback. The player starts by powering the lab using energy options associated with the energy simulation variables in Table 3. The main goal is to evolve and release the creature on the island. Within 30 game days, the player will be prompted to engage with the other characters in the research facility. This timeline enables players to explore and make a series of decisions that occur each day to interpret the feedback for the impact of their decisions. The game's turn-based nature forces players to investigate and develop their observation and deduction skills for the scientific method. To begin decision-making, the player observes Illumi's health and changes the energy options and room configurations. Interactions with the other characters such as scientist, engineer, and librarian can further enhance the energy knowledge, while players can observe and learn about the impact of the energy choice on Illumi's health status. Players would also maintain information on the tablet that tracks decisions made as the game progressed. Based on the pre-simulation results and unique ids of energy options, the tablet displays updated energy performances of EUI,

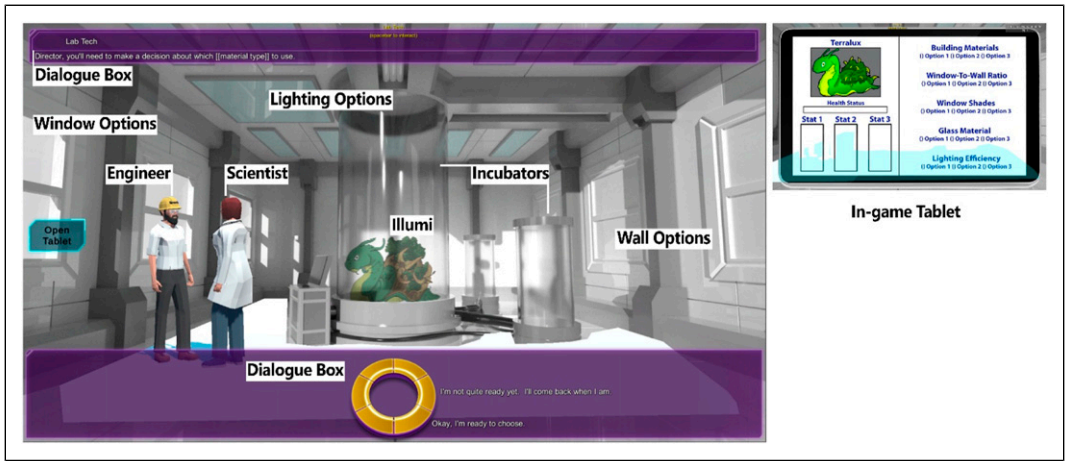


Figure 5. The game interface (left) and the dashboard (right).

energy cost, and CO2 emissions. These indicators enable players to monitor how their energy choices affect environmental conditions.

Interactions with other characters and dialogue information also provide players with textual guidance to iterate each energy option further. Sub-objectives given by other characters enable players to gather wide-range information regarding how each energy option, such as lighting choices, material changes, and window resizing, are associated with environmental footprints. When all information has been explored for a game day, players will indicate to the lab assistant that they are ready to finalize their decision. The tablet will show the decision made and the information the player has gathered with a hypothesized prediction for how the creature's health will turn out the next day. The game ends at the end of either 30 days when the creature is fully evolved or earlier if the player fails to accommodate the creature's health each day. When the creature fails to thrive, a unique learning opportunity exists for the player to identify the cause with the option to try again, which is important not to discourage further interaction with the game.[43] The alignment with the curriculum modules further supports players in comparing the choices made against the accompanying course materials to build more motivation and engagement in subsequent playthroughs.

Table 5. Sample game dialogues.

Category	Example dialogues
Wall/Roof materials	You can change the materials of the building elements, such as walls or roofs to improve the overall energy performance. Please test the three options below to help the creature!
Window-to-wall ratio	Did you know that the window size impacts the energy needed to heat or cool the building? Test the three options below to figure out which one is better for energy consumption in your laboratory!
Window shades	Windows are essential for view and fresh air but remember that the laboratory gets very hot in the summer. All of that extra heat will require air conditioning! Test the shading devices below to see if you can improve the energy performance of your laboratory in the summer
Glass materials	Will changing the glass material in your windows help you use less energy and save the creature? Please choose from the three options below
Lighting efficiency	It is time to choose lightbulbs for your laboratory! Please test the three options below to find which one will lower energy usage

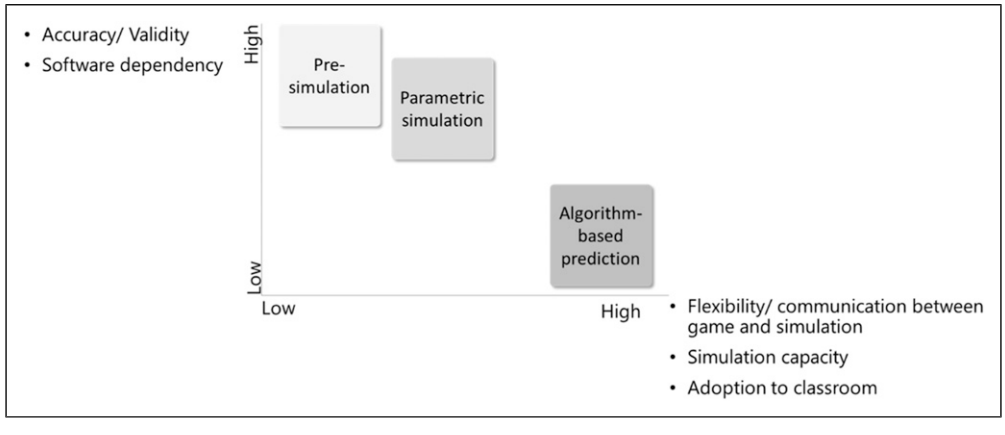


Figure 6. Comparison of the integration methods between energy modeling, simulation, and game development.

Besides, the research explored and compared three methods to integrate BIM and energy simulations to feed a large number simulation data into the education games in the context of rural middle school classrooms with limited technology accessibility (Figure 6). The main criteria for the comparison includes (i) accuracy/validity of simulation results, the dependency of energy modeling and simulation, flexibility in game and simulation development, simulation capacity to handle a wide range of game behavior and energy options, feasibility for the classroom adoption, etc.

The research carried out pre-simulation using BIM-based plug-ins and cloud simulations built on the EnergyPlus engine to gain the validity of simulation results. Pre-simulation needs to identify the complete set of energy simulation scenarios, considering the type and number of energy options and game behaviors. Once the simulation was completed, changes in the curriculum and the game options were not allowed.

Conclusion

Green building educational opportunities are rare outside engineering and design programs in higher education. However, the concept and principles of energy-efficient and net-zero buildings are aligned with the goal of formal science learning. We aspired to develop a process for serious game development that can facilitate seamless communication and data flow across three domains of curriculum development, energy modeling/simulation, and game development. The research attempted to take advantage of immerging technology of parametric BIM modeling, energy simulations, and game development. To move towards comprehensive serious game development, the research started with a 2.5 d mini-game, *Illumi's World*. It produced a theoretical and empirical foundation for the full Net-Zero Game.

Illumi's World was an outcome of the co-development of curriculum, building modeling, energy simulation, and game development. More often, educational games exist outside of formal curriculum materials and are used as "add-ons" (e.g., PhET interactives) in the classroom when teachers choose to access a free educational game or simulation that may only be partially aligned with the unit learning goals. We observed that the linear and top-down process could deteriorate flexibility in revisions, content alignments, and data exchange in serious game development and the game's adoption to classrooms. In Illumi's World, the game design fits within the curriculum structure as it was specifically designed to work within the EYE unit. Illumi's World serves to facilitate learning as well as provide a means to assess student learning through their iteration of choices within the game. This is a novel educational intervention that is new to the field of science education in which an education game was designed to embed within the existing curriculum.

One of the significant implications is the necessity of systematic data management between energy modeling/simulation and serious game. The data process from a simulation into a game proved challenging. Games provide abstracted data in the form of a system that guides players to meet objectives, whereas simulations work with a multitude of data that enables users to self-direct and manipulate key details. This difference in scope of data uses led to a need to combine variables and select key representative results from the simulations to present within the game's interface. With our intended learner group, middle schoolers, and the need to encourage the exploration of choices, we had to consider multiple game elements and energy options combined with the data. The story and scene were helpful in setting the players up to role-play, but the dialogue and the feedback system were necessary to communicate the data. Identifying a balanced design without losing the integrity of the simulation data is an aspect we aim to explore further as we shift our efforts to the larger Net-Zero game.

BIM adoption in serious games still faces several challenges, particularly when considering game adoption in classrooms. For instance, real-time communication between BIM and games is available only when the classroom can access full software licenses. Data exchange and communication between BIM and games remain limited. Game development platforms have limited access to BIM data, and BIM tools have insufficient features for game development and vice versa. In addition, our mini-game relied on prerendered

images from BIM and an image-swapping function to update game scenes. This led to an inefficient process that worked but required a series of renderings to load with decisions, forcing the design not to use real-time feedback. However, our study into how we enabled energy simulation data to work in a game framework has provided insights into how we should address a larger-scale game.

When considering the broad curriculum scopes, various energy options in the game can exponentially increase the cost of energy simulations. Pre-simulations based on the predefined game scenario can prevent flexible game scenarios and curriculum revisions. Reversely, it can cause complex changes in game design after the energy simulation runs. We anticipate that algorithm-based prediction may mediate the limitation of top-down and pre-simulation. It would eliminate the potential software license issues when distributing the serious game to the classrooms. The full Net-Zero Game will further investigate this simulation method.

The game's development is still in its early stage and needs feedback from the teachers and students to prove our approach. The under-developing research phase is the usability test of the mini-game, planned to be performed at rural middle school districts in Southern Missouri. Within the U.S., rural school districts are underserved populations for STEM education. They do not have ready access to green buildings in their communities. This challenge presents crucial equity issues as rural student expectations towards energy workforces increase, such as the gas, oil, wind, water, and solar energy industries. The described game is expected to bring knowledge about energy flow, human energy use, and green buildings to the rural students.

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